

PsExec Hunt Lab

Analyze SMB traffic in a PCAP file using Wireshark to identify PsExec lateral movement, compromised systems, user credentials, and administrative shares.

Category:

[Network Forensics](#)

Tactics:

[Execution](#)[Defense Evasion](#)[Discovery](#)[Lateral Movement](#)

Tool:

[Wireshark](#)

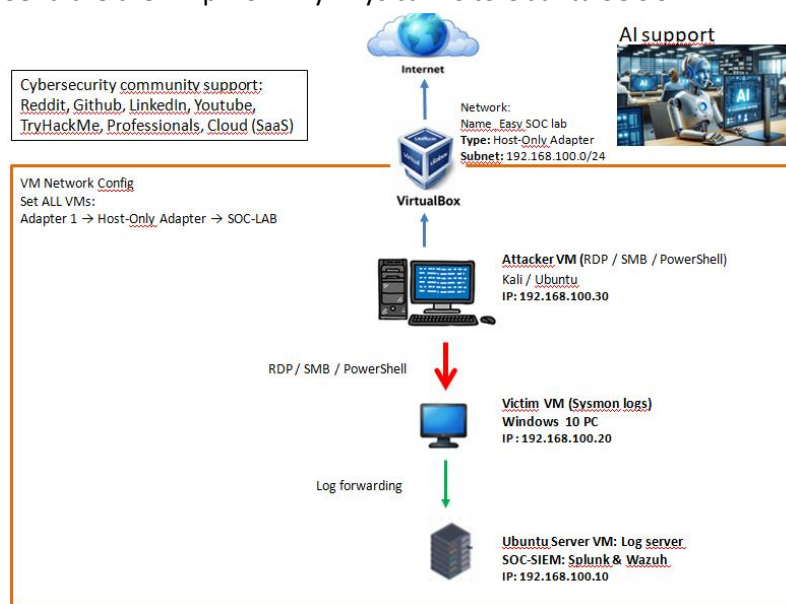
Using my Lab as sandbox to open the Files and check the files with Virus total (no malware found)

An alert from the Intrusion Detection System (IDS) flagged suspicious lateral movement activity involving PsExec. This indicates potential unauthorized access and movement across the network. As a SOC Analyst, your task is to investigate the provided PCAP file to trace the attacker's activities. Identify their entry point, the machines targeted, the extent of the breach, and any critical indicators that reveal their tactics and objectives within the compromised environment. On the windows PC

1-download the zip 143-psexec-hunt

3- open my SOC LAB

Send the the 2 zip from my Physical PC to Ubuntu SOC SIEM VM



Step 1: Configure on host

1. Shut down the VM

2. Go to **Settings → Shared Folders**
3. Add a folder:
 - Folder Path → choose a host directory containing your (ZIP 143-psexec-hunt)
 - Folder Name → psexerchuntlab
 - ☒ Auto-mount
 - ☒ Make Permanent

Step 2: Boot VM and mount

(create this directory if not exist)

```
sudo mkdir -p /mnt/shared
```

```
sudo mount -t vboxsf psexerchuntlab /mnt/shared
```

Now your ZIP will be available at:

```
jojo@jojo:/mnt/shared$xxx.zip
```

Step 3: Fix permissions

```
sudo usermod -aG vboxsf $USER
```

check :

```
ls -l /mnt/shared/xxx.zip
```

```
-rwxrwxrwx 1 root root 332 Apr 18 11:04 /mnt/sharedxxx.zip
```

Breakdown:

- `rw-` → owner permissions
- `r--` → group permissions
- `last ---` → others
- `root vboxsf` → owner and group

4) Test access explicitly

```
cat /mnt/shared/xxx.zip > /dev/null
```

IF

- Works → permissions OK
- Fails → still blocked

`-rwxrwxrwx` → **777 permissions**

How to Install Wireshark on Ubuntu 24.04

<https://www.youtube.com/watch?v=8dA9c4hbslo>

first update

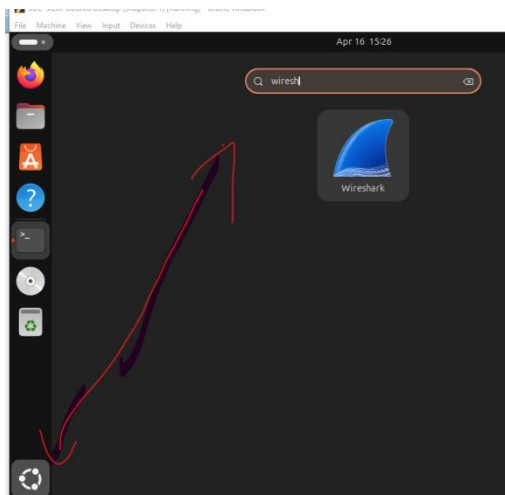
```
jojo@jojo:~$ sudo apt update
```

1-ubuntu standard installation

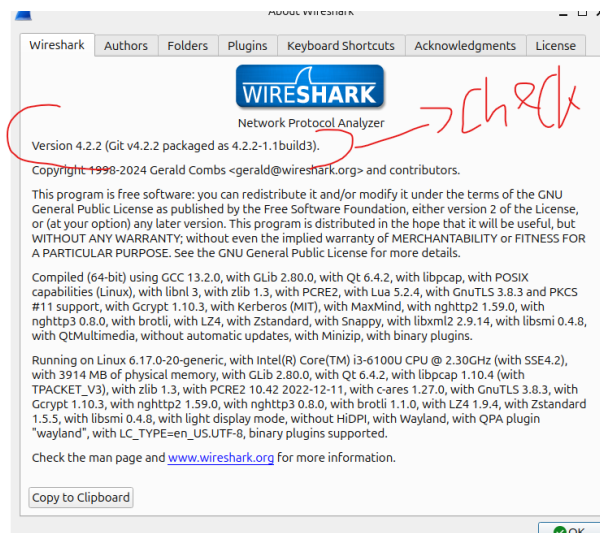
```
jojo@jojo:~$ sudo apt install wireshark -y
```

A- when asked : let the default [NO] and press enter

Run Wireshark: in application menu



Check the Version maybe not the later version



If the version is not current you can remove

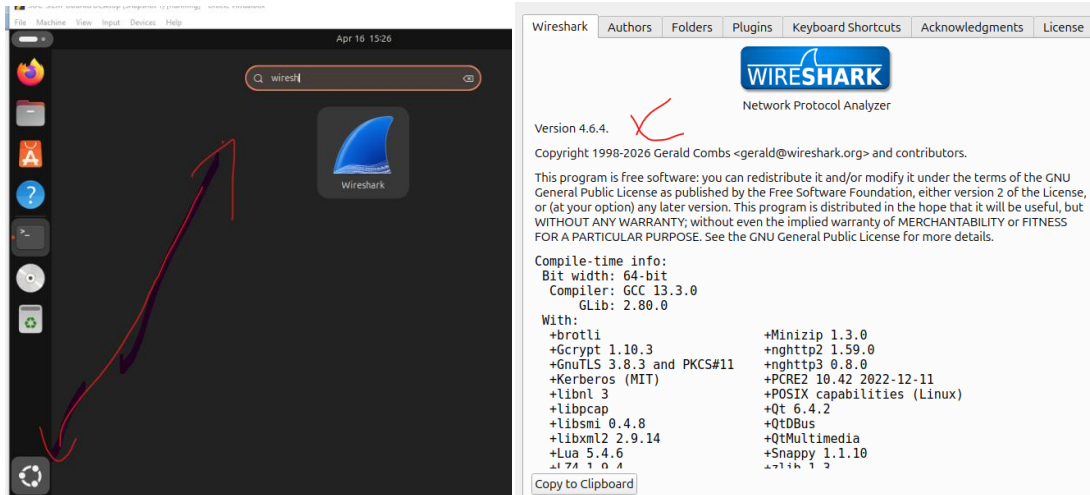
```
Sudo apt remove wireshark -y
```

To install wireshark from the wireshark repository (it will include the latest version).

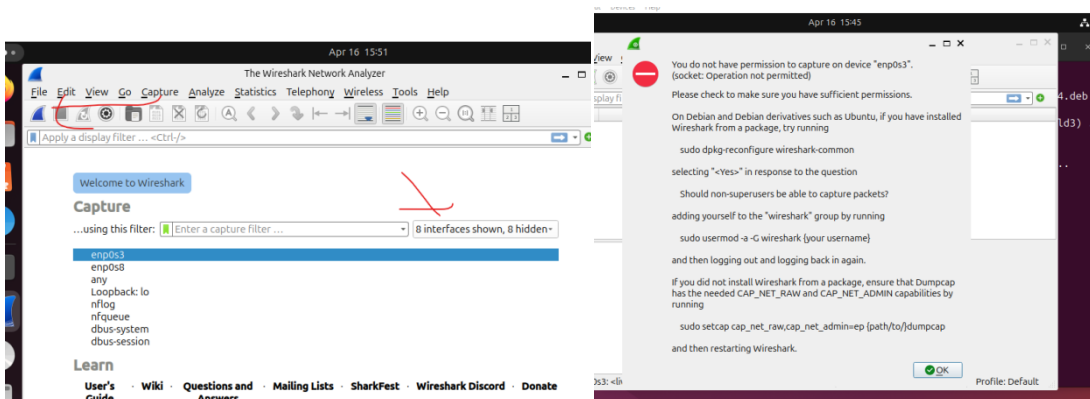
1-jojo@jojo:~\$ sudo add-apt-repository ppa:wireshark-dev/stable

2-jojo@jojo:~\$ sudo apt install wireshark -y

open Wireshark: in application menu and check the version

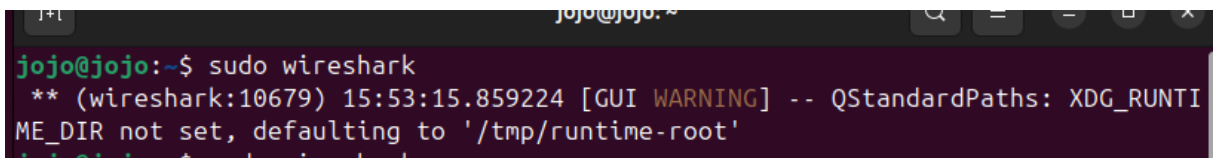


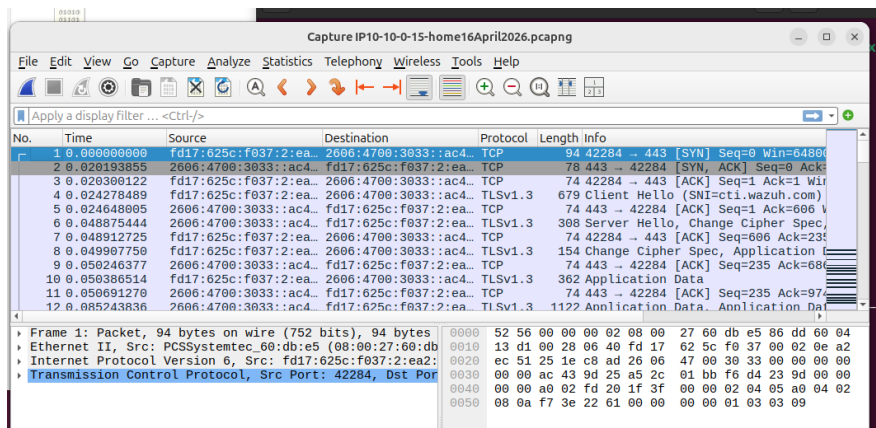
If having this error



Launch wireshark from Commande

jojo@jojo:~\$ sudo wireshark





Important: 2 days of search and work in html supported with AI coding , also use of <https://tryhackme.com/room/aithreatmodelling>)

I created with support of AI my one internat use cheat sheet for Wireshark SQL, called wireshark Analyst Bible (complete Filter & analysis guide) in html :200+ Filters in 15 categories

Now ready to investigate the PCAP file.

Q1

Solved : 9645

To effectively trace the **attacker's** activities within our network, can you **identify the IP address** of the machine from which the attacker initially gained access?

1-Check External IP sending SYN packetsInitial connection attempt

list all IP from the traffic

GUI Method: Statistics → Endpoints → IPv4 tab

Or

GUI Method: Statistics → Conversation → IPv4 tab

Check the IP with higher Packet and byte

IP 10.0.0.130 has the most packets

Wireshark - Endpoints - psexec-hunt.pcapng

Endpoint Settings

- ☐ Name resolution
- ☐ Display raw data
- ☐ Hide aggregated
- ☐ Limit to display filter
- Copy
- Map
- Protocol
 - ☐ Bluetooth
 - ☐ BPv7
 - ☐ DCCP
 - ☐ DNP 3.0
 - ☒ Ethernet
 - ☐ FC
- Filter list for specific type

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City
10.0.0.1	241	27 kB	41	12 kB	200	15 kB		
10.0.0.130	40,040	9 MB	20,374	5 MB	19,666	4 MB		
10.0.0.131	1,768	2 MB	631	473 kB	1,137	1 MB		
10.0.0.132	200	15 kB	200	15 kB	0	0 bytes		
10.0.0.133	38,284	7 MB	19,047	3 MB	19,237	4 MB		
10.0.0.254	1	342 bytes	1	342 bytes	0	0 bytes		
10.0.0.255	32	4 kB	0	0 bytes	32	4 kB		
224.0.0.22	7	422 bytes	0	0 bytes	7	422 bytes		
224.0.0.252	1	72 bytes	0	0 bytes	1	72 bytes		
239.255.255.250	14	10 kB	0	0 bytes	14	10 kB		

In the conversation IP 10.0.0.130 and IP 10.0.0.133 have 4MB the most packets

Wireshark - Conversations - psexec-hunt.pcapng

Conversation Settings

- ☐ Name resolution
- ☐ Absolute start time
- ☐ Display raw data
- ☐ Limit to display filter
- Copy
- Follow Stream...
- Graph...
- I/O Graphs
- Protocol
 - ☐ Bluetooth
 - ☐ BPv7
 - ☐ DCCP
- Filter list for specific type
- Help

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B
10.0.0.1	10.0.0.255	27	2 kB	1	27	2 kB
10.0.0.1	239.255.255.250	14	10 kB	6	14	10 kB
10.0.0.130	10.0.0.131	1,755	2 MB	9	1,136	1 MB
10.0.0.130	10.0.0.133	38,284	7 MB	7	19,237	4 MB
10.0.0.130	10.0.0.255	1	243 bytes	8	1	243 bytes
10.0.0.131	10.0.0.255	4	972 bytes	5	4	972 bytes
10.0.0.131	224.0.0.22	7	422 bytes	3	7	422 bytes
10.0.0.131	224.0.0.252	1	72 bytes	4	1	72 bytes
10.0.0.132	10.0.0.1	200	15 kB	0	200	15 kB
10.0.0.254	10.0.0.131	1	342 bytes	2	1	342 bytes

In conclusion the IP-src is 10.0.0.130

Look for failed connections (RST packets)

tcp.flags.reset == 1

Wireshark - Packet List - psexec-hunt.pcapng

Filter: tcp.flags.reset == 1

No.	Time	Source	Destination	Protocol	Length	Info
38	510.858100073	10.0.0.130	10.0.0.133	TCP	60	49700 → 49669 [RST, ACK] Seq=1419 Ac
38	510.858259882	10.0.0.130	10.0.0.133	TCP	60	49699 → 135 [RST, ACK] Seq=329 Ac=2
38	521.565896161	10.0.0.130	10.0.0.133	TCP	60	49696 → 445 [RST, ACK] Seq=2533606 A
38	534.444996618	10.0.0.130	10.0.0.131	TCP	60	49701 → 445 [RST, ACK] Seq=1041 Ac=
39	536.970047062	10.0.0.130	10.0.0.131	TCP	60	49704 → 135 [RST, ACK] Seq=329 Ac=2
39	536.970056269	10.0.0.130	10.0.0.131	TCP	60	49705 → 49669 [RST, ACK] Seq=2283 Ac
39	546.006580719	10.0.0.130	10.0.0.131	TCP	60	49703 → 445 [RST, ACK] Seq=386058 Ac
39	548.472381186	10.0.0.130	10.0.0.131	TCP	60	49708 → 135 [RST, ACK] Seq=329 Ac=2
39	548.472388119	10.0.0.130	10.0.0.131	TCP	60	49709 → 49669 [RST, ACK] Seq=2363 Ac
39	552.524562779	10.0.0.130	10.0.0.131	TCP	60	49707 → 445 [RST, ACK] Seq=386397 Ac
40	555.050803304	10.0.0.130	10.0.0.131	TCP	60	49713 → 49669 [RST, ACK] Seq=2363 Ac
40	555.051200223	10.0.0.130	10.0.0.131	TCP	60	49712 → 135 [RST, ACK] Seq=329 Ac=2

Frame 38484: Packet, 60 bytes on wire (480 bits), 60 byte captured (480 bits) on interface 0
 Ethernet II, Src: VMware_06:cc:75 (08:0c:29:06:cc:75), Dst: 08:00:00:00:00:00
 Internet Protocol Version 4, Src: 10.0.0.130, Dst: 10.0.0.133
 Transmission Control Protocol, Src Port: 49700, Dst Port: 49669

Look for PsExec Specifically

SMB traffic (PsExec uses SMB)

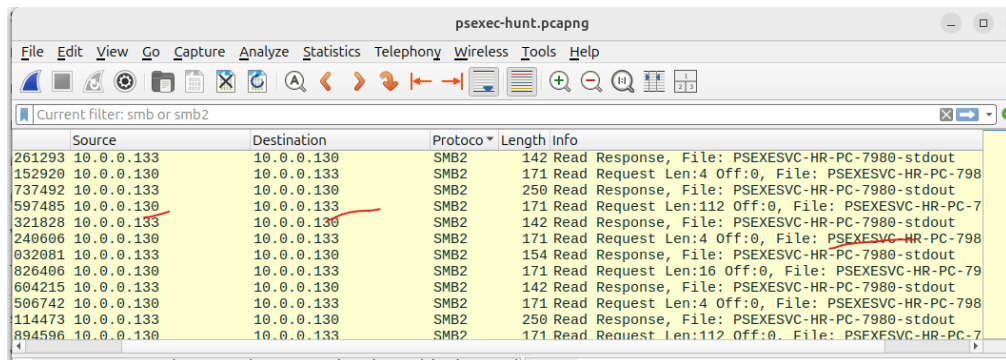
smb or smb2

PsExec service name in traffic

smb2 and frame contains "PSEXESVC"

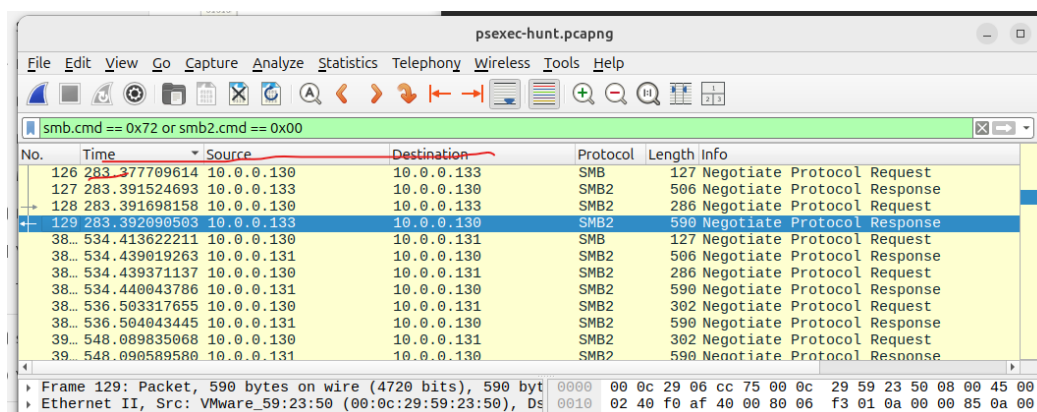
SMB on port 445

tcp.port == 445



Source	Destination	Protocol	Length	Info
261293 10.0.0.133	10.0.0.130	SMB2	142	Read Response, File: PSEXESVC-HR-PC-7980-stdout
152920 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:4 Off:0, File: PSEXESVC-HR-PC-798
737492 10.0.0.133	10.0.0.130	SMB2	250	Read Response, File: PSEXESVC-HR-PC-7980-stdout
597485 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:112 Off:0, File: PSEXESVC-HR-PC-7
321828 10.0.0.133	10.0.0.130	SMB2	142	Read Response, File: PSEXESVC-HR-PC-7980-stdout
240606 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:4 Off:0, File: PSEXESVC-HR-PC-798
932081 10.0.0.133	10.0.0.130	SMB2	154	Read Response, File: PSEXESVC-HR-PC-7980-stdout
826406 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:16 Off:0, File: PSEXESVC-HR-PC-79
604215 10.0.0.133	10.0.0.130	SMB2	142	Read Response, File: PSEXESVC-HR-PC-7980-stdout
506742 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:4 Off:0, File: PSEXESVC-HR-PC-798
114473 10.0.0.133	10.0.0.130	SMB2	250	Read Response, File: PSEXESVC-HR-PC-7980-stdout
894596 10.0.0.130	10.0.0.133	SMB2	171	Read Request Len:112 Off:0, File: PSEXESVC-HR-PC-7

We can confirm by Focusing on the initial SMB (Server Message Block) negotiation traffic by using the filter `smb.cmd == 0x72` or `smb2.cmd == 0x00` to Identify the first Source IP address connected via SMB Protocol.



No.	Time	Source	Destination	Protocol	Length	Info
126	283.377709614	10.0.0.130	10.0.0.133	SMB	127	Negotiate Protocol Request
127	283.391524693	10.0.0.133	10.0.0.130	SMB2	506	Negotiate Protocol Response
128	283.391698158	10.0.0.130	10.0.0.133	SMB2	286	Negotiate Protocol Request
129	283.392090503	10.0.0.133	10.0.0.130	SMB2	590	Negotiate Protocol Response
38...	534.413622211	10.0.0.130	10.0.0.131	SMB	127	Negotiate Protocol Request
38...	534.439019263	10.0.0.131	10.0.0.130	SMB2	506	Negotiate Protocol Response
38...	534.439371137	10.0.0.131	10.0.0.131	SMB2	286	Negotiate Protocol Request
38...	534.440043786	10.0.0.130	10.0.0.130	SMB2	590	Negotiate Protocol Response
38...	536.503317655	10.0.0.130	10.0.0.131	SMB2	302	Negotiate Protocol Request
38...	536.504043445	10.0.0.131	10.0.0.130	SMB2	590	Negotiate Protocol Response
39...	548.089835068	10.0.0.130	10.0.0.131	SMB2	302	Negotiate Protocol Request
39...	548.090589580	10.0.0.131	10.0.0.130	SMB2	590	Negotiate Protocol Response

Response= 10.0.0.130

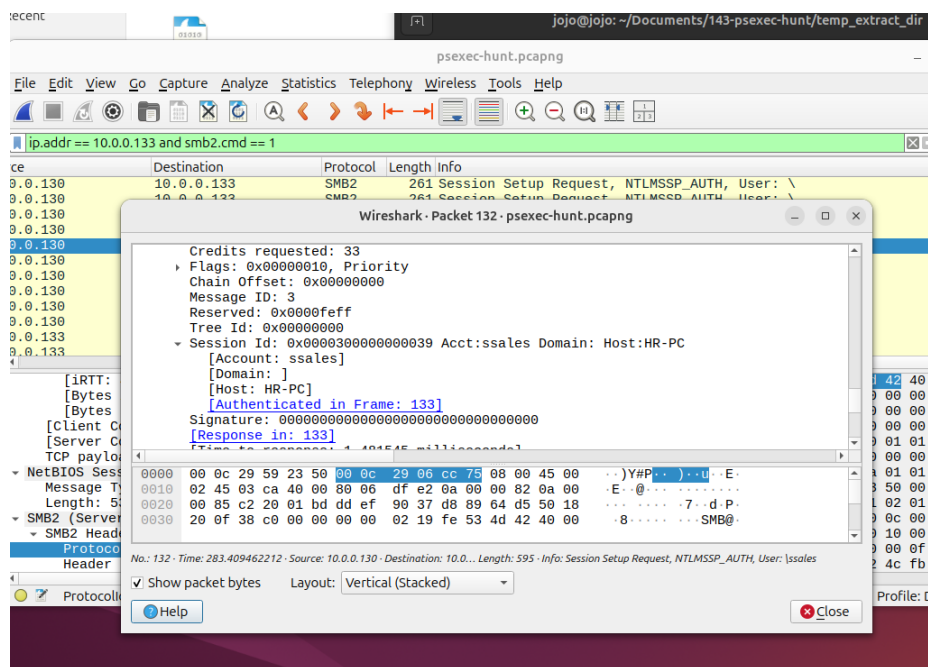
Q2

Solved : 9085

To fully understand the extent of the breach, can you determine the machine's hostname to which the attacker first pivoted?

From the previous question we know the first destination IP was 10.0.0.133

ip.addr == 10.0.0.133 and smb2.cmd == 1

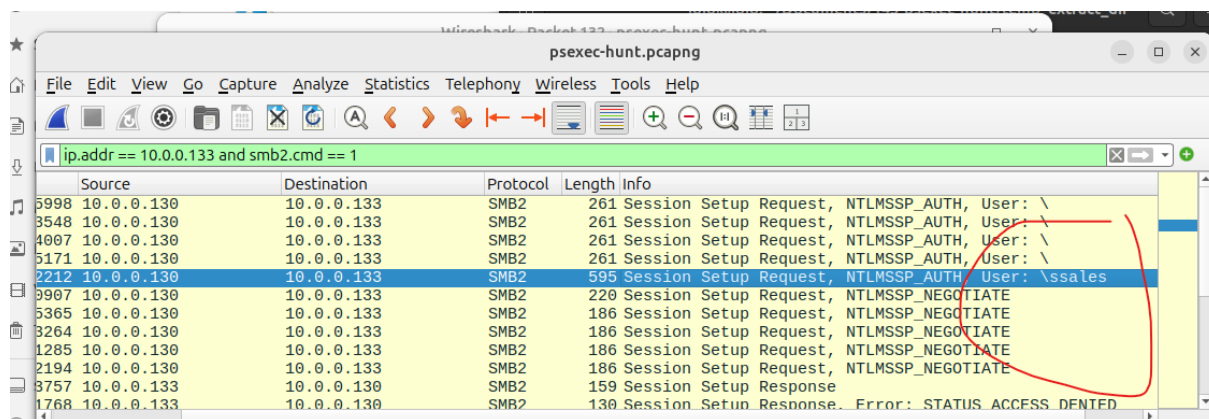


Response: sales-pc

Q3

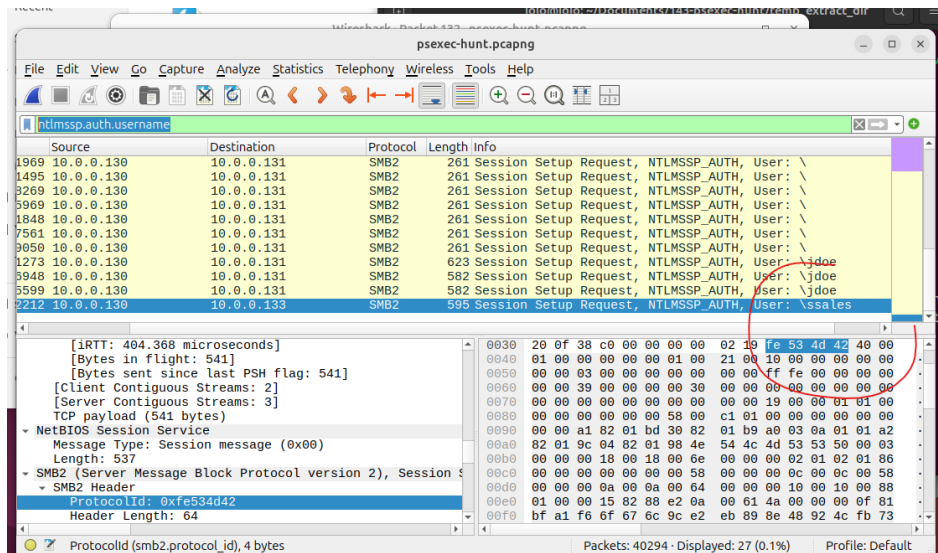
Solved : 9003

Knowing the username of the account the attacker used for authentication will give us insights into the extent of the breach. What is the username utilized by the attacker for authentication?



Or filter

ntlmssp.auth.username



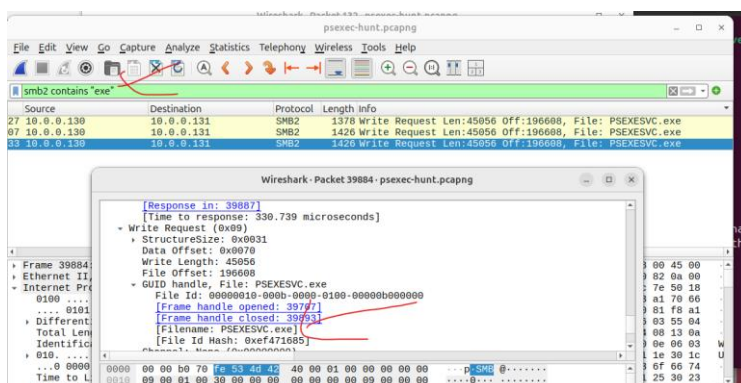
Response : ssales

Q4

Solved : 8895

After figuring out how the attacker moved within our network, we need to know what they did on the target machine. What's the name of the service executable the attacker set up on the target?

smb2 contains "exe"



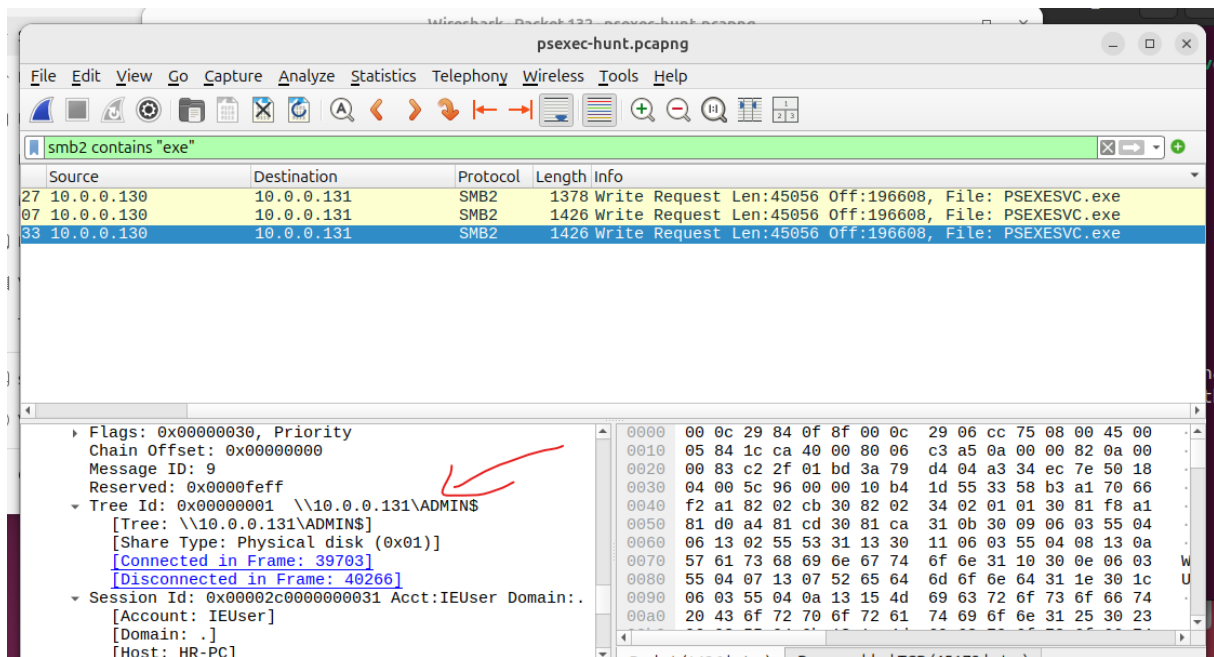
RESPONSE: PSEXESVC.exe

Q5

Solved : 8884

We need to know how the attacker installed the service on the compromised machine to understand the attacker's lateral movement tactics. This can help identify other affected systems. Which network share was used by PsExec to install the service on the target machine?

smb2 and ip.src == 10.0.0.133



Response: ADMIN\$

Q6

Solved : 8849

We must identify the network share used to communicate between the two machines. Which network share did PsExec use for communication?

smb2 and ip.src == 10.0.0.133

Common PsExec Shares

Share

Purpose

ADMIN\$ Maps to C:\Windows — PsExec uploads PSEXESVC.exe here

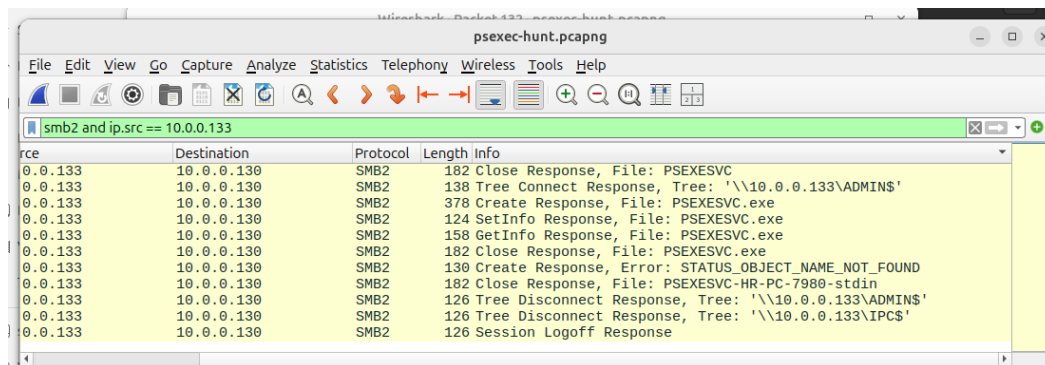
IPC\$ Used for RPC calls to Service Control Manager

C\$ Full C drive access — sometimes used as alternative

SMB2

└─ Tree Connect Request

└─ Path: \\TARGET\IPC\$ □ ← Communication share HERE



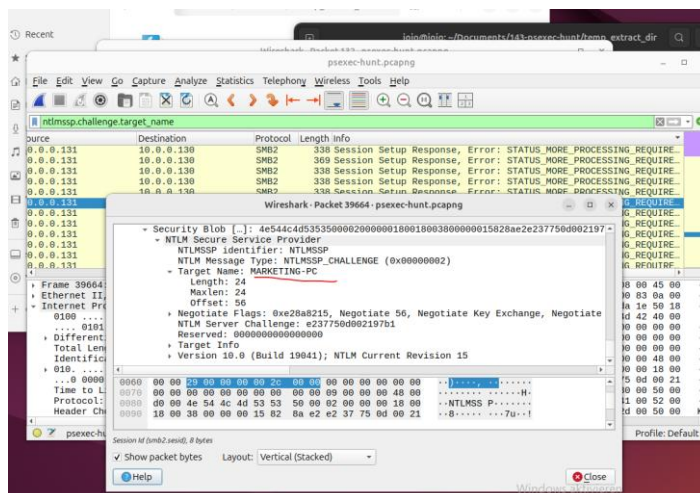
Response: IPC\$

Q7

Solved : 8747

Now that we have a clearer picture of the attacker's activities on the compromised machine, it's important to identify any further lateral movement. What is the hostname of the second machine the attacker targeted to pivot within our network?

ip.addr == 10.0.0.131 and smb2.cmd == 1



Response: marketing-PC

Summary or the Project

Threat Hunt: Unmasking PsExec Lateral Movement from Raw Network Traffic Hands-On SOC Lab | April 2026 | CyberDefenders Blue Team CTF Wireshark · SMB/NTLM Analysis · VirtualBox · Ubuntu 24.04

Built a personal SOC lab from scratch and hunted a simulated attacker through raw PCAP data — no hints, no shortcuts. Using advanced Wireshark filtering and protocol-level analysis, reconstructed the complete attack chain of a PsExec-based intrusion across multiple hosts.

What I uncovered:

- ☐ Traced the attacker's entry point — pinpointed source IP via SMB negotiation traffic and conversation analysis
- ☐ Mapped the full pivot path — identified two compromised machines (`sales-pc` → `marketing-pc`) and the stolen credential used to move laterally
- ☒ Exposed the attack mechanism — detected `PSEXESVC.exe` deployment and the exact shares abused (`ADMIN$` for execution, `IPC$` for C2 communication)
- ☐ Tagged to MITRE ATT&CK — Execution · Lateral Movement · Defense Evasion · Discovery

Impact: Demonstrated ability to independently detect, trace, and document a multi-stage intrusion using only network evidence — a core SOC Tier 1/2 skill.

For On linkedin

<https://cyberdefenders.org/blueteam-ctf-challenges/psexec-hunt/>

🔍 I just hunted an attacker through raw network traffic — and it was intense.

As part of my SOC analyst journey, I completed a hands-on threat hunting lab today: tracking a PsExec lateral movement attack from scratch using only a PCAP file and Wireshark.

No alerts handed to me. No pre-filtered logs. Just raw network traffic and the question: *where did the attacker go?*

Here's what I uncovered ☐

☐ Entry point — identified the attacker's source IP by digging into SMB negotiation traffic and endpoint conversations

☐ Stolen credential — extracted the compromised username via NTLMSSP authentication analysis

☒ Attack mechanism — detected `PSEXESVC.exe` being dropped remotely and traced the exact admin shares abused (`ADMIN$` for execution, `IPC$` for communication)

☐ Full pivot path — mapped the attacker's movement across two machines before they could go further

☐ Mapped to MITRE ATT&CK framework:

- T1570 · Lateral Tool Transfer
- T1021.002 · Lateral Movement via SMB/Windows Admin Shares
- T1078 · Valid Accounts (*stolen credentials*)
- T1569.002 · System Services: Service Execution
- T1083 · File and Directory Discovery

All of this in a lab I built myself — VirtualBox + Ubuntu SOC environment, Wireshark, and a CyberDefenders Blue Team CTF scenario.

The biggest lesson? Attackers leave traces. You just have to know where to look.

If you're breaking into cybersecurity, hands-on labs like this are worth more than any certification alone. Build the lab. Get your hands dirty. Do the work.

□ CyberDefenders | #SOC #ThreatHunting #BlueTeam #NetworkForensics #Wireshark
#MITRE #ATTACKFramework #LateralMovement #PsExec #Cybersecurity #CareerInCyber